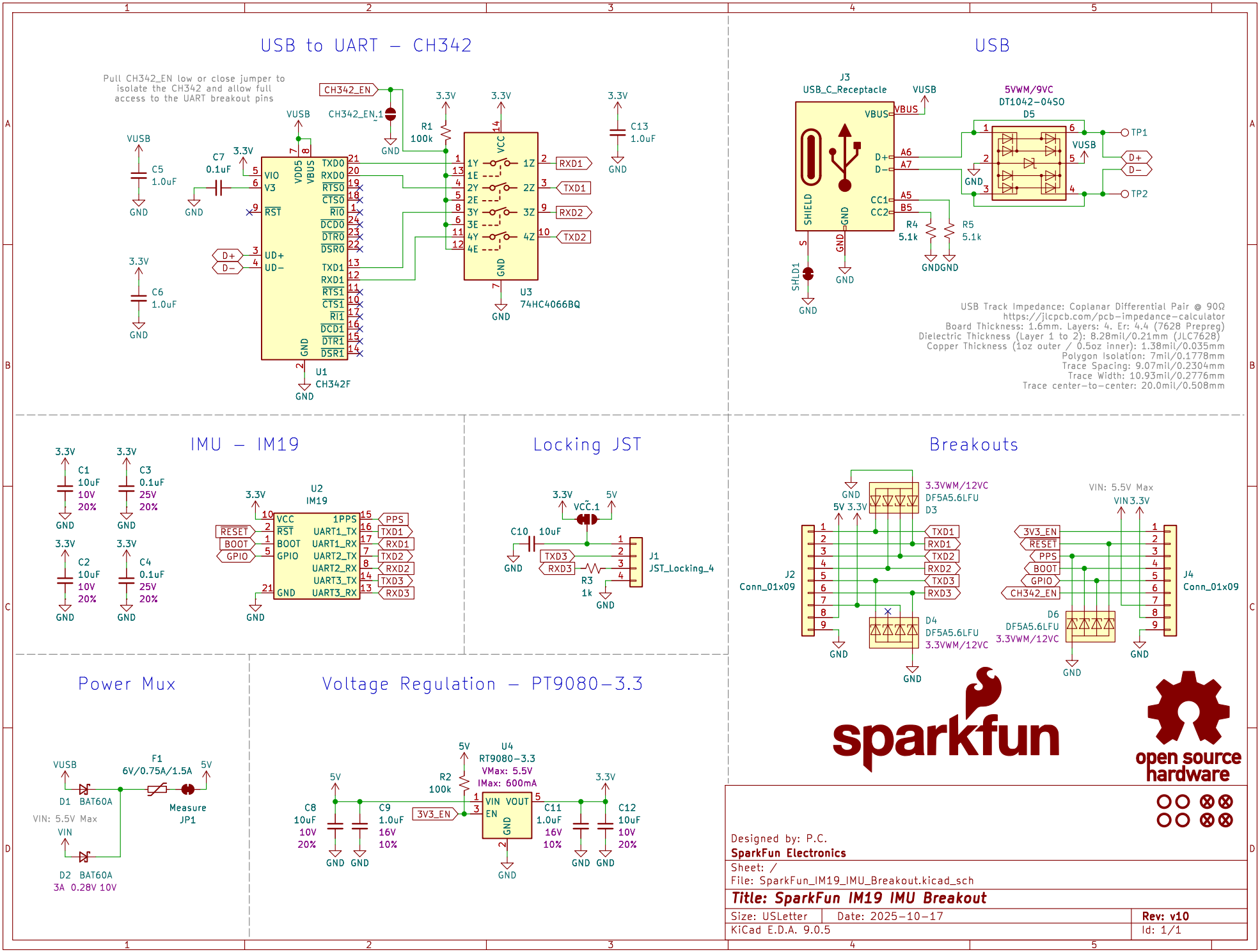
USB to UART – CH342
Pull CH342_EN low or close jumper to isolate the CH342 and allow full access to the UART breakout pins
CH342_EN
CH342_EN_1
VUSB
C5 1.0uF
C7 0.1uF
3.3V
C6 1.0uF
R1 100k
C13 1.0uF
U1 CH342F
U3 74HC4066BQ
RXD1
TXD1
RXD2
TXD2
USB
J3 USB_C_Receptacle
5VWM/9VC DT1042-04SO D5
TP1
TP2
D+
D–
R4 5.1k
R5 5.1k
SHLD1
USB Track Impedance: Coplanar Differential Pair @ 90Ω
https://jlcpcb.com/pcb-impedance-calculator
Board Thickness: 1.6mm. Layers: 4. Er: 4.4 (7628 Prepreg)
Dielectric Thickness (Layer 1 to 2): 8.28mil/0.21mm (JLC7628)
Copper Thickness (1oz outer / 0.5oz inner): 1.38mil/0.035mm
Polygon Isolation: 7mil/0.1778mm
Trace Spacing: 9.07mil/0.2304mm
Trace Width: 10.93mil/0.2776mm
Trace center-to-center: 20.0mil/0.508mm
IMU – IM19
C1 10uF 10V 20%
C2 10uF 10V 20%
C3 0.1uF 25V 20%
C4 0.1uF 25V 20%
U2 IM19
RESET
BOOT
GPIO
PPS
Locking JST
C10 10uF
VCC_1
J1 JST_Locking_4
R3 1k
Breakouts
J2 Conn_01x09
J4 Conn_01x09
3.3VWM/12VC DF5A5.6LFU D3
D4 DF5A5.6LFU 3.3VWM/12VC
D6 DF5A5.6LFU 3.3VWM/12VC
VIN: 5.5V Max
3V3_EN
CH342_EN
Power Mux
F1 6V/0.75A/1.5A
D1 BAT60A
D2 BAT60A 3A 0.28V 10V
Measure JP1
Voltage Regulation – PT9080-3.3
U4 RT9080-3.3
VMax: 5.5V
IMax: 600mA
R2 100k
C8 10uF 10V 20%
C9 1.0uF 16V 10%
C11 1.0uF 16V 10%
C12 10uF 10V 20%
Designed by: P.C.
SparkFun Electronics
Sheet: /
File: SparkFun_IM19_IMU_Breakout.kicad_sch
Title: SparkFun IM19 IMU Breakout
Size: USLetter
Date: 2025-10-17
Rev: v10
KiCad E.D.A. 9.0.5
Id: 1/1